

Difficulty-aware score generation for piano sight-reading

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ABSTRACT

Sight-reading is a core skill in music education. It refers to the ability to play a written piece of music correctly the first time it is seen. Developing this skill requires frequent practice with completely new musical excerpts that match the difficulty level of the student. However, creating new sight-reading exercises at a specific difficulty level requires significant time and expert knowledge. As a result, students and teachers often rely on pieces from the existing piano literature, even though sight-reading exams typically use compositions written specifically for the exam. Generative music systems provide a promising approach for creating new sight-reading material with explicit control over performance difficulty. In this work, we frame the creation of sight-reading exercises as a symbolic music generation task that produces piano scores with controllable difficulty. Existing approaches typically rely on control tokens to guide generation, but we show that this strategy does not result in piano scores with reliably controlled difficulty. To address this issue, we introduce an auxiliary difficulty prediction objective using synthetic difficulty labels produced by an expert-based system, enabling scalable training. Our method improves difficulty conditioning accuracy from 69.3% to 92.9% compared to a baseline that conditions generation solely on difficulty control tokens, and reduces mean squared error from 0.30 to 0.09. A user study with expert pianists shows that the generated scores are rated comparable or superior to human-written material in readability and naturalness, while maintaining appropriate playability across difficulty levels. These results represent a step toward the generation of music exercises for a variety of educational applications.

1. Introduction

Sight-reading refers to performing a piece of music at first sight, without prior preparation (Spencer, 2011). Unlike other musical skills, effective sight-reading practice requires constant exposure to previously unseen material, since learners must not have played or heard the music before (Wristen, 2005). Sight-reading is a core skill in music education and a standard component of piano exams and curricula worldwide (Associated Board of the Royal Schools of Music, 2024; Trinity College London, 2021; Yamaha Corporation, 2025). In these contexts, assessment typically relies on short, newly composed pieces rather than works from the existing repertoire.

The development of sight-reading skills strongly depends on frequent practice with unfamiliar material that is carefully matched to the current ability of the learner (Farley, 2014; Wristen, 2005). In practice, creating a large amount of original, level-appropriate exercises is costly and difficult for teachers and students. As a consequence, practice often relies on a small set of pre-existing exercises drawn from published piano literature (Bovin, 2021). Learners therefore frequently practice with material

that is either too easy, leading to boredom, or too difficult, which limits effective learning and progression (Pike & Shoemaker, 2013; Wristen, 2005). This well-defined educational setting makes sight-reading practice a concrete and promising scenario for automatic exercise generation. This is especially true if the generated material supports gradual progression through careful difficulty control, which music pedagogy often emphasizes (Hayward & Eastlund Gromko, 2009; Pike & Shoemaker, 2013). However, previous research has mainly focused on the analysis and assessment of sight-reading skills, rather than on the generation of level-appropriate practice material (Cheng et al., 2008; Elliott, 1982; Huang & Lerch, 2019).

These pedagogical requirements for generating progressive material motivate computational approaches that can produce new practice material while controlling key educational constraints. In practice, the challenge addressed in this work is the automatic generation of piano sight-reading exercises that are both musically coherent and controllable in terms of performance difficulty. In this paper, we treat this task as a special case of personalized generation. We use the term *personalized* to denote difficulty-targeted practice: given a target difficulty level for

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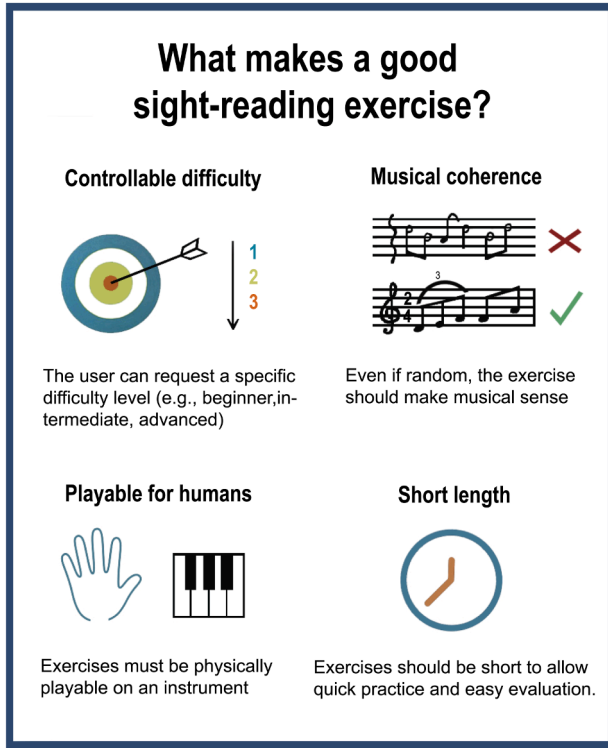


Fig. 1. Effective sight-reading exercises must be brief, coherent, playable, and appropriate for the level of the player. Our system explicitly controls difficulty while implicitly addressing the other constraints.

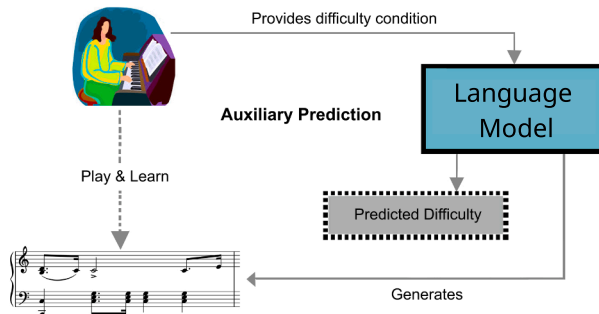


Fig. 2. Overview of the proposed system for generating piano exercises with controllable difficulty. The model is trained autoregressively on symbolic music and conditioned on target difficulty levels. An auxiliary difficulty prediction task is included during training to encourage representations aligned with difficulty, although it is not used at generation time.

a specific learner, the system generates new exercises at that level. Personalization is therefore limited to explicit difficulty control, as set by a teacher, a curriculum, or an estimate derived from recently practiced repertoire. We do not model other aspects such as stylistic preferences or specific technical goals.

As a first step toward personalized generation, we explore how recent advances in piano difficulty prediction can support conditioning generative music models on target difficulty levels. In addition to difficulty control, effective sight-reading exercises, as illustrated in Fig. 1, must exhibit musical coherence: they should be readable by humans, playable on the instrument, and concise enough to fit typical practice routines.

In particular, we propose a symbolic music generation system that produces short and readable piano scores in MusicXML format, condi-

tioned on a target difficulty level. Previous research has controlled autoregressive models by adding conditioning tokens at the beginning of the sequence (Li & Liang, 2021). However, when the target attribute is performance difficulty, token-based conditioning remains challenging in practice, as such approaches can suffer from conditioning collapse and provide limited control over the overall difficulty of the generated scores, an effect reported in multiple domains not related with music generation (Dathathri et al., 2019; Lee et al., 2024; Liu et al., 2024; Skopek et al., 2023). Following prior work in other domains (Li et al., 2020; Yu et al., 2024), we introduce a lightweight auxiliary difficulty prediction objective to support difficulty-controlled generation. Fig. 2 provides an overview of the proposed system. In addition, because large-scale expert difficulty annotation is impractical and expensive, our approach relies on synthetic difficulty labels predicted by an expert system trained on prior expert-annotated data (Ramoneda et al., 2024c).

Recent advances in the generation of symbolic music with language models have shown strong capabilities to produce coherent and stylistically controlled compositions (Huang et al., 2019; Huang & Yang, 2020; Ma et al., 2021). Related work has also demonstrated the impact of intelligent generative systems in creative and educational domains beyond music, including visual media (Aydam & El Abbadi, 2025) and broader educational settings (Mijwil et al., 2023). However, most previous work has focused on Musical Instrument Digital Interface (MIDI) generation (Hu et al., 2022b; Huang et al., 2019; Huang & Yang, 2020), which is often not ideal when the output must be read and practiced as standard musical notation. Unlike Music Extensible Markup Language (MusicXML), MIDI files are primarily designed for sound synthesis and do not explicitly encode notation structure. To generate exercises that are immediately usable by students and teachers, we directly produce MusicXML scores (Good, 2021), a widely adopted open standard supported by educators, notation software, and digital archives. Using a previously introduced tokenization method (Suzuki, 2021), we finetune pretrained large language models on MusicXML data to enable high-quality, legible score generation.

Our main contributions are:

- We formulate difficulty-aware sight-reading exercise generation as a controlled symbolic generation problem in MusicXML, directly targeting human-readable scores required in real music education settings.
- We show that standard prompt-based conditioning is often insufficient to control the overall difficulty of generated scores, and identify conditioning collapse as a key limitation in this context.
- We leverage synthetic difficulty labels predicted by an expert system to enable scalable difficulty-aware conditioning, and combine them with a lightweight auxiliary difficulty prediction loss to regularize generation toward target difficulty levels.
- We adapt pre-trained language models to symbolic music and validate the proposed approach through objective metrics and a user study with expert pianists, demonstrating improved controllability and educational relevance.

Our goal is to support scalable creation of short, readable exercises that match a target difficulty level. To facilitate replication and future work, we will release code and models,¹ and provide a companion page with examples².

2. Literature review

This section reviews the research foundational to our personalized sight-reading system, focusing on score representation, difficulty prediction and generation.

¹ At: <https://github.com/PRamoneda/sight-reading-generation>

² At: <https://sight25.github.io>

2.1. Score representations

Symbolic music representations are a key component in most computational music systems. Early approaches focused on MIDI or piano roll formats due to their simplicity and availability (Hsiao et al., 2021; Huang et al., 2019; Huang & Yang, 2020). However, these formats lack important information such as notational structure, visual layout, or expressive markings, essential for tasks such as music education, performance, or score editing.

To overcome the limitations of MIDI and piano roll, recent works adopt richer notation formats such as MusicXML, which preserve both musical semantics and visual aspects of the score (Good, 2021). Other approaches use discrete token-based representations tailored for sequential models. For example, Score Transformer proposes a linearized token format for the generation of piano music (Suzuki, 2021), while Measure by Measure introduces a hierarchical grid structure based on side bars (Yan & Duan, 2024). ABC notation, a compact text-based format, offers a simpler alternative that supports fast composition modeling and is increasingly used in folk and choral music generation, as seen in models such as FolkRNN (Sturm et al., 2016), Tunesformer (Wu et al., 2023b) or DeepChoir (Wu et al., 2023a). In addition to symbolic processing tasks, transcription tasks such as Optical Music Recognition or audio-to-score conversion also benefit from richer notation formats, including Humdrum (Huron, 1995) and MusicXML variants (Yan & Duan, 2024), which provide structured and expressive representations to align audio, visual, and symbolic data.

While ABC-based representations are effective for certain musical styles, they are less suited for large-scale educational applications due to the lack of extensive, high-quality datasets and the limited expressiveness of the format for piano notation. Moreover, conversion between symbolic formats (e.g., ABC, MIDI, and MusicXML) is not lossless and can introduce errors in rhythmic structure, voice assignment, or notation layout, which are critical for sight-reading practice. Among the different representations, we adopt the one proposed by Suzuki (2023), as it is based on MusicXML and aligns with our goal of generating readable and editable sheet music. MusicXML is one of the most widely used formats in music education, notation software, and digital archives, making it a suitable choice for data-intensive machine learning tasks. Our tokenization method preserves structural details such as measures, voices, and staves, as well as essential musical information including note names, durations, ties, stems, beams, and articulations. By converting these musical elements into linear token sequences, we facilitate efficient score modeling while retaining musical coherence.

2.2. Difficulty prediction

Predicting musical difficulty has been widely studied in the context of music education and performance analysis. Because collecting large-scale expert annotations is costly and time-consuming, many computational approaches aim to estimate difficulty automatically from symbolic or performance-related features.

Organizing educational material into steps of increasing difficulty is essential for effective learning. Playing an instrument involves not only reading the score but also body movement, sound production, and interpretative principles tied to specific performance traditions (Cook, 1999). As a result, performance difficulty can be understood as the effort required by a pianist to meet the expressive, technical, and notational demands of a passage (Ramoneda et al., 2024a). Since objective criteria are limited, difficulty assessments often involve subjectivity, leading to disagreements among educators and grading systems (Wareham, 1967). Nevertheless, rankings from music boards, conservatories, composers, and publishers reveal consistent global trends (Ramoneda et al., 2024a,c; Zhang et al., 2023).

Early approaches relied on hand-crafted features extracted from symbolic scores, such as note density, pitch range, and articulation, combined with classical or rule-based machine learning models (Chiu & Chen, 2012; Sébastien et al., 2012). Later studies incorporated performance-based features, including finger movement statistics, to better capture technical demands (Nakamura et al., 2014; Nakamura & Sagayama, 2015). Recent work explores deep learning models trained on expert-annotated MusicXML scores (Ramoneda et al., 2024a,c), as well as image (Ramoneda et al., 2023) and audio data (Ramoneda et al., 2024b), achieving strong performance but reduced interpretability.

To improve explainability, RubricNet (Ramoneda et al., 2024c) introduced a set of interpretable descriptors inspired by pedagogical rubrics, capturing aspects such as pitch range, entropy, rhythmic density, hand displacement, and structural repetitiveness. These descriptors provide a structured and pedagogically grounded representation of piano performance difficulty and form the basis of several recent difficulty estimation approaches.

2.3. Related generative work on difficulty control

Several studies have also addressed difficulty-controlled score rearrangement. Nakamura and Yoshii (Nakamura & Yoshii, 2018) formulate piano reduction as a statistical optimization problem balancing difficulty and fidelity via fingering models. Gover and Zewi (Gover & Zewi, 2022) adopt a MIDI translation approach to generate piano arrangements at different playing levels. Suzuki (2023) propose a notation-to-notation transformer model that modifies existing piano scores while preserving expressive marks. Unlike these approaches, we generate novel, sight-reading exercises from scratch, conditioned on interpretable difficulty descriptors. This design enables fine-grained control and better pedagogical alignment, bridging the gap between music informatics and music education.

More broadly, applied intelligent systems research spans a wide range of methodological foundations, including formal algebraic analysis, differential equation modeling, and reliability-oriented system design. Although these directions are outside the specific scope of symbolic music generation, they provide complementary perspectives on system-level modeling and analysis in applied intelligent systems (Abed and Khalil, 2023; C. et al., 2025; Mohammed et al., 2025).

3. Methodology

Our method for generating difficulty-aware symbolic music consists of four stages: (1) synthetic difficulty labelling (2) self-supervised pre-training using next-token prediction, (3) difficulty conditioning with various prompt strategies, and (4) refinement of the conditioning with an auxiliary loss. More generally, the use of auxiliary objectives and system-level constraints is common in other areas of applied modeling and optimization, including reliability-oriented system design (Abed et al., 2025).

3.1. Synthetic difficulty labeling

Difficulty is modeled in this work as an explicit control variable that enables difficulty-aware symbolic music generation. Since large symbolic music corpora rarely include reliable expert difficulty annotations, we adopt an automatic labeling strategy that provides consistent difficulty supervision at scale.

Difficulty estimation is performed using a lightweight probabilistic classifier based on interpretable pedagogical descriptors, following the RubricNet framework (Ramoneda et al., 2024c). These descriptors capture musically and technically relevant properties of piano scores, including pitch range, pitch entropy, rhythmic density, hand

Table 1

Difficulty prediction performance on the CIPI dataset (Ramoneda et al., 2024a). We report nine-class classification accuracy (Acc-9; higher is better) and mean squared error (MSE; lower is better) between predicted and reference difficulty levels. Values in parentheses indicate variability across runs or splits as reported in the corresponding studies.

	CIPI Dataset Acc-9	MSE
argnn variant (Ramoneda et al., 2024a)	32.6 (2.8)	2.1 (0.2)
virtuoso model (Ramoneda et al., 2024a)	35.2 (7.3)	2.1 (0.2)
pitch-feat model (Ramoneda et al., 2024a)	32.2 (5.9)	1.9 (0.2)
deep ensemble (Ramoneda et al., 2024a)	39.5 (3.4)	1.1 (0.2)
RubricNet (Ramoneda et al., 2024c)	41.4 (3.1)	1.7 (0.5)
Gaussian Naive Bayes (ours)	38.7 (4.2)	1.8 (0.3)

displacement, average pitch, and structural repetitiveness. All descriptors are computed directly from symbolic MusicXML data.

We use a probability-calibrated Gaussian Naive Bayes classifier (John & Langley, 1995) trained on expert-annotated datasets from prior work (Ramoneda et al., 2024a,c). These datasets are based on the CIPI dataset (Ramoneda et al., 2024a) and follow the Henle Verlag difficulty scale.³ This scale is defined by editors with a research background in musicology and comes from one of the most well-known publishers of piano music. We choose this model for its simplicity, interpretability, and robustness, which are useful for large-scale synthetic labeling.

Given a descriptor vector f , the classifier outputs a probability distribution over difficulty classes Y . The predicted difficulty $\hat{y}$ is obtained as

$$\hat{y} = \arg \max_{y \in Y} P(y | f). \quad (1)$$

In addition to the predicted class, we retain the maximum posterior probability $\max_y P(y | f)$ as a confidence estimate. To reduce label noise, we discard samples with low confidence by removing the least confident 25% of predictions. This filtering step yields a cleaner and more consistent pool of symbolic excerpts labeled by difficulty, which is used throughout conditioning and training.

Table 1 reports the performance of representative difficulty prediction models on the CIPI dataset, including the Gaussian Naive Bayes baseline adopted in this work. Although more complex models achieve higher accuracy, the selected classifier provides explicit probabilistic outputs that allow us to associate each predicted label with a confidence score. This confidence estimate is used to filter low-confidence predictions, resulting in a more consistent set of synthetic difficulty labels. As an additional qualitative sanity check, an external piano teacher with more than 40 years of teaching experience reviewed 100 automatically labeled excerpts and identified 21 cases as misaligned with the assigned difficulty level. While this check does not constitute a formal evaluation, it suggests that the majority of retained labels are pedagogically plausible and suitable for training difficulty-aware generative models at scale.

3.2. Self-supervised pretraining

We pretrain a decoder-only Transformer using an autoregressive objective on symbolic music tokens, as shown in Fig. 3. The model uses causal masking (masked self-attention) to prevent access to future tokens during training. The representation we adopt is based on Suzuki (2023), as introduced in Section 2.1. Each score is linearized into a

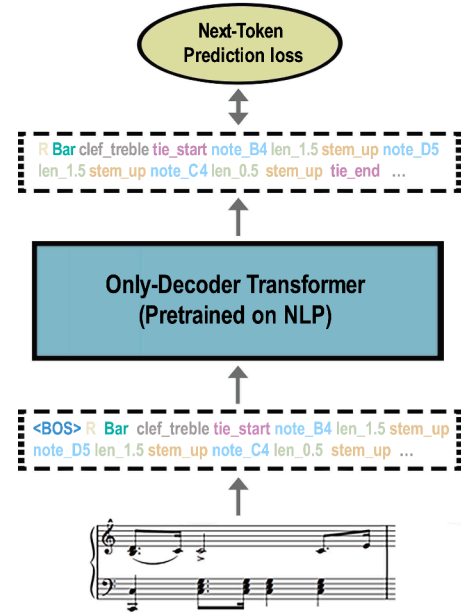


Fig. 3. Self-supervised pretraining of a decoder-only Transformer on large-scale MusicXML data using next-token prediction. This stage allows the model to learn general score syntax and musical structure, and provides a pretrained representation for subsequent difficulty conditioning.

sequence of discrete tokens (e.g., note names, durations, articulations, markers), enabling the use of standard language modeling techniques.

Let $x = (x_1, x_2, \dots, x_T)$ be a sequence of tokens. The model estimates the probability of the sequence via causal factorization:

$$P(x) = \prod_{t=1}^T P(x_t | x_{<t}; \theta_{lm}), \quad (2)$$

where θ_{lm} are the model parameters.

We minimize the negative log-likelihood of the correct token at each timestep, resulting in the standard cross-entropy loss:

$$\mathcal{L}_{CE} = - \sum_{t=1}^T \log P(x_t | x_{<t}; \theta_{lm}). \quad (3)$$

We experiment with two families of architectures pre-trained in text: LLaMA3 models (Grattafiori et al., 2024) and the HuggingFace Smol v2 (Allal et al., 2024) series. To enable efficient fine-tuning, we use low-rank adaptation (LoRA) (Hu et al., 2022a), a parameter-efficient technique that injects trainable low-rank matrices into the frozen weights of the pre-trained model.

Formally, for a weight matrix $W \in \mathbb{R}^{d \times k}$, LoRA introduces two trainable matrices $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times k}$, with $r \ll \min(d, k)$. During fine-tuning, the weight is updated as:

$$W' = W + \alpha AB, \quad (4)$$

where α is a scaling factor. This reduces the number of trainable parameters while preserving performance.

To condition the model on difficulty, we adopt a prepending strategy, where a natural language prompt indicating the target difficulty is placed at the beginning of each sequence, as shown in Fig. 4. This allows the model to adjust its generation according to the desired difficulty level while preserving coherence and musical structure.

The models used in this phase are those pre-trained in the self-supervised stage and finetuned using LoRA, as described in Section 3.2. This approach enables efficient adaptation to the MusicXML structure while retaining the language knowledge of the original model. Thanks to this, we can use natural language prompts (e.g., "Easy") to guide the generation process.

³ Available at: <https://www.henle.de/en/Levels-of-Difficulty/>

Table 2

Examples of the four prompt types used to encode difficulty during fine-tuning. The strategies vary in length, explicitness, and semantic richness, ranging from simple zero-shot labels to more detailed chain-of-thought descriptions.

Type	Prompt Content
diff	Easy
diff_cot	To compose a piano piece, we start by defining its characteristics. An Easy piece should have a clear melody, limited movement, and a steady rhythm. A Mid piece introduces a wider pitch range, moderate hand movement, and rhythmic variety. An Advanced piece demands technical skill, with complex passages, fast sequences, and wide hand displacement. Now, generate a Easy piano composition in MusicXML format, ensuring structure and musical coherence.
feats	Easy: 0.21 0.19 0.35 0.28 0.50 0.45 0.60 0.58 0.33 0.31 0.27 0.29
feats_cot	We will generate a Easy piano piece. First, we analyze its musical descriptors. Pitch entropy shows pitch variety. RH: 0.21, LH: 0.19. Higher = more diverse notes. Pitch range is the distance between lowest and highest notes. RH: 0.35, LH: 0.28. Avg. pitch gives the central register. RH: 0.50, LH: 0.45. Higher = brighter. Displacement rate reflects hand movement intensity. RH: 0.60, LH: 0.58. Avg. IOI is the time between note onsets. RH: 0.33, LH: 0.31. Lower = denser rhythm. Pitch set LZ shows structural complexity and repetitiveness. RH: 0.27, LH: 0.29. Now generate a coherent and structured piece in MusicXML.

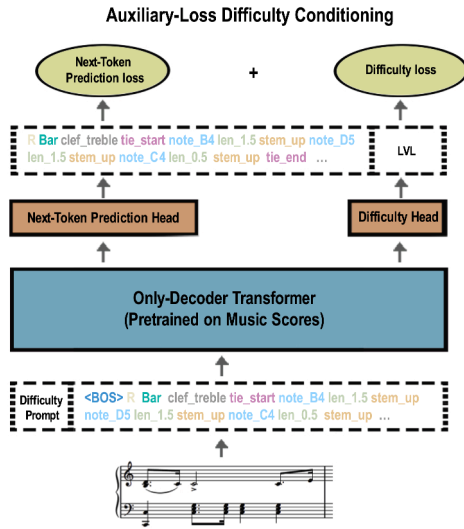


Fig. 4. Difficulty conditioning setup. A difficulty prompt is prepended to the MusicXML token sequence, and an auxiliary difficulty prediction loss is applied during training. Empirically, this additional objective improves adherence to the target difficulty.

To minimize context length, self-supervised pretraining was performed with an extended vocabulary that included MusicXML-specific symbols, incorporating the tokenization vocabulary introduced in Section 2.1. However, we did not modify the original natural language vocabulary. During the difficulty conditioning stage, we finetune both vocabularies jointly: the music vocabulary and the natural language vocabulary.

3.3. Difficulty conditioning

Given the sequence x and a prompt p , the conditioning is modeled as:

$$P(x | p) = \prod_{t=1}^T P(x_t | p, x_{<t}; \theta_{lm}), \quad (5)$$

where θ_{lm} are the parameters of the only-decoder transformer and the loss remains the standard autoregressive cross-entropy loss. In the computation of $\mathcal{L}_{lm}$, we ignore the prompt p .

For each training fragment x , we compute a difficulty label $d(x) \in \mathbb{N}_0$ and optionally features $f(x) \in \mathbb{R}^n$, as described in Section 2.2. These labels are *synthetic* and obtained using the Gaussian Naive Bayes model introduced in Section 2.2. They are then used to build the prompt $p(x)=g(f(x), d(x), t)$, where $t \in \{\text{diff}, \text{diff_cot}, \text{feat}, \text{feat_cot}\}$ denotes the prompt type used for fine-tuning.

The *diff* format is a minimal, zero-shot prompt containing only the difficulty label $d(x)$ (e.g., Advanced). The *diff_cot* format extends this with a natural-language description of what difficulty implies, following a chain-of-thought strategy to enhance reasoning (Wei et al., 2022). The *feat* format encodes only the normalized extracted features $f(x)$, while *feat_cot* combines $f(x)$ with a verbalized explanation, also following a chain-of-thought strategy. Table 2 provides examples of these prompt formats.

3.4. Multi-objective optimization with an auxiliary loss

Difficulty is a complex and subjective concept and does not align well with the cross-entropy loss described in the previous section. In practice, generating with difficulty conditioning using only the main autoregressive objective can cause *conditioning collapse*, where the model learns to ignore the conditioning signal, either fully or partially. This issue has been observed in other generation tasks that involve style, sentiment, or structure. To address this, previous work has explored solutions such as auxiliary objectives (Lapacz et al., 2025; Li et al., 2020; Yu et al., 2024; Zhu et al., 2023), latent space interpolation (Lee et al., 2024), constraint-based refinement (Skopek et al., 2023), and inference-time control (Dathathri et al., 2019). These studies show that autoregressive loss alone is not enough to ensure that the model respects complex conditioning, especially when dense supervision is lacking. Following this line, we introduce an auxiliary task that explicitly models the difficulty. The effectiveness of using auxiliary objectives to improve conditioning has been clearly demonstrated by prior studies in other applications (Lapacz et al., 2025; Li et al., 2020; Yu et al., 2024), and in our case, this additional objective helps the model better match the intended difficulty level.

For clarity, we summarize the training pipeline and the interaction between the autoregressive objective and the auxiliary difficulty prediction task in Algorithm 1.

Algorithm 1 Training step with auxiliary difficulty prediction.

Require: Token sequence x , difficulty label d , prompt p , weight β

- 1: $x' \leftarrow \text{concat}(p, x)$
- 2: $H \leftarrow \text{Transformer}(x')$
- 3: $L_{CE} \leftarrow \text{CrossEntropy}(\text{NTHead}(H), x)$
- 4: $h_{\text{final}} \leftarrow H_{\text{END}}$
- 5: $L_{\text{diff}} \leftarrow \text{CrossEntropy}(\text{DiffHead}(h_{\text{final}}), d)$
- 6: $L_{\text{total}} \leftarrow L_{CE} + \beta \cdot \text{stopgrad}_{\theta_{LM}}(L_{\text{diff}})$
- 7: Update all parameters using gradients of L_{total}

Initially, the input sequence x is preceded by a natural language prompt p , which encodes the desired difficulty level, as outlined in Section 3.3. Second, we add an auxiliary prediction head at the final token position (denoted END) that classifies the target difficulty d . This head operates on the final hidden state of the sequence and is trained to predict the intended difficulty class.

Let d be the target difficulty label (e.g., easy, medium, advanced). We define the auxiliary classification loss as:

$$\mathcal{L}_{\text{diff}} = -\log P(d \mid x; \theta_{\text{diff}}), \quad (6)$$

where θ_{diff} are the parameters of the difficulty classification head.

The total training loss is a weighted combination of the autoregressive and auxiliary objectives:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \beta \cdot \mathcal{L}_{\text{diff}}, \quad (7)$$

where β is a scalar that controls the influence of the auxiliary loss.

To reduce the risk of shortcuts—such as the model copying the difficulty label from the prompt into the final classification head—we detach the gradient flow from the auxiliary classification loss $\mathcal{L}_{\text{diff}}$ into the parameters θ_{LM} of the language model. In other words, during back-propagation, only the classification head parameters θ_{diff} are updated by $\mathcal{L}_{\text{diff}}$, while θ_{LM} remains unaffected by this loss: $\nabla_{\theta_{\text{LM}}} \mathcal{L}_{\text{diff}} = 0$, and $\nabla_{\theta_{\text{diff}}} \mathcal{L}_{\text{diff}} \neq 0$. This design choice aims to discourage trivial solutions based on surface patterns in the prompt and instead promote the encoding of difficulty information within the generated content. Although the auxiliary head does not directly influence the generation, it provides a weak supervision signal that may help to reinforce the consistency between the output and the intended difficulty level.

4. Experimental setup

4.1. Data

We use the PDMX dataset, recently introduced by Long et al. (2025), which contains more than 200,000 public domain MusicXML scores collected from the MuseScore platform. PDMX is the largest copyright-free MusicXML dataset available and includes rich metadata such as user tags and ratings, and we use it as our main training corpus.⁴

From this collection, we extract 41,170 piano pieces with exactly two staves and high community ratings. We segment each piece into 16-bar fragments and estimate their difficulty using the synthetic difficulty labeling procedure described in Section 3. Based on these predictions, fragments are grouped into three difficulty classes: Easy (level 0), Medium (level 1), and Advanced (level 2 or higher).

We emphasize that difficulty labels are not provided by the PDMX dataset itself, but are automatically inferred using a lightweight, interpretable classifier trained on expert-annotated data from prior work. These synthetic labels are used exclusively for difficulty-aware fine-tuning and conditioning, and do not affect the self-supervised learning of musical structure.

Although the PDMX dataset is skewed toward easier fragments (61.5% Easy, 37.0% Medium, 12.1% Advanced), we balance samples across difficulty classes during fine-tuning to avoid bias in the conditional generation model. Similarly, for evaluation of difficulty control, we generate an equal number of samples per class to ensure fair comparison 3.

We divide the 41,170 piano scores into 80% for training and 20% for validation. To increase diversity and improve robustness to key changes, we apply automatic transpositions of -6 and $+6$ semitones to the training set using MuseScore (MuseScore Development Team, 2024), adjusting the key signature accordingly. To avoid introducing noisy or rare notation, we discard transposed samples that result in double sharps or double flats. This produces a musically consistent augmented training set containing 505,838 unique piano fragments and more than 659 million symbolic tokens, which is used during self-supervised pretraining.

To evaluate generalization beyond the PDMX distribution, we include three additional datasets during self-supervised pretraining and

Table 3

Summary of the datasets used in this work, reporting the number of pieces and the corresponding number of raw MusicXML tokens after tokenization. These datasets are used for self-supervised pretraining, validation, and out-of-distribution evaluation of the MusicXML generative model.

Dataset	Pieces	Tokens (raw)
PDMX Train	33,962	45,764,701
PDMX Validation	7,176	10,026,819
PDMX Train (augmented)	505,838	659,061,903
Polish Classical	1,532	4,581,980
Craig Sapp Classical	852	5,406,551
Internal (J-Pop)	1,061	6,192,391

validation: a collection of Polish classical scores, the Craig Sapp Classical dataset, and a private dataset of Japanese pop music. These datasets are used exclusively to assess robustness and out-of-distribution generalization of the symbolic music model, and are not used for difficulty labeling or difficulty-conditioned fine-tuning, which is performed only on PDMX due to copyright constraints.

4.2. Training details

We tokenize all symbolic music using the vocabulary introduced in Section 2.1, based on a MusicXML structure (Suzuki, 2021). To reduce noise and sparsity, we discard tokens that appear less than 50 times in the training set, resulting in a compact vocabulary that still captures the musical variety of PDMX. All experiments use LoRA adapters with rank 64 and alpha 128 to enable parameter-efficient finetuning with reduced memory usage. Initial tests with other common configurations showed similar results. Experiments are conducted on a single NVIDIA A100 GPU, using gradient accumulation when needed to fit large batches in memory.

We use the AdamW optimizer in all training steps and train until convergence. Early stopping is applied every two epochs, based on validation performance. For self-supervised pretraining, we monitor the validation loss on the validation split of PDMX. For conditioning strategies, we evaluate difficulty prediction accuracy on 1,000 generated samples. During pretraining, we use a cosine learning rate scheduler with a minimum value of 0.001 and a batch size of 128. For difficulty conditioning, we used a fixed learning rate of 0.0001 and a batch size of 16, as larger sizes did not show clear benefit. We set $\beta = 0.1$ for the auxiliary prediction loss.

5. Results

5.1. Self-supervised finetuning

Table 4 reports the validation loss in four benchmark datasets after the self-supervised pretraining stage in PDMX Train. All models were trained in the same symbolic music corpus using the cross-entropy objective described in Section 3.2. We include models of different sizes and pretraining strategies to compare generalization, convergence speed, and adaptation to symbolic music. In particular, we selected two families of pre-trained models (LLaMA3 (Grattafiori et al., 2024) and Smol (Al-lal et al., 2024)) to explore how language pretraining impacts symbolic music modeling, and we included GPT-2 as a baseline widely used in prior midi generation models (Pasquier et al., 2025; Thickstun et al., 2024).

The best overall performance is achieved by LLaMA 8B, which obtains the lowest validation loss across all datasets with only 40k training steps. This suggests that large-scale models pre-trained on text can efficiently adapt to symbolic music with relatively few updates. In contrast, GPT-2, trained from scratch without prior NLP pretraining, requires 120k steps and yields the highest validation loss, confirming the

⁴ We train directly on MusicXML, the most widely used open format among users, since alternative formats such as MIDI or ABC lack essential notation details (e.g., pitch spelling, beam grouping, and voice separation) and have limited reliable conversion tools to MusicXML.

Table 4

Model performance measured as validation cross-entropy loss (lower is better) during the self-supervised pretraining stage on symbolic music data. *NLP Pretrain* indicates whether the model was pretrained on language data. *Steps* shows the number of training steps needed for convergence.

Model	Params	Pret.	Steps	PDMX Val	Polish	Classical	J-Pop
GPT-2	100M	✗	120k	0.38	0.64	0.66	1.21
LLaMA3	1B	✓	50k	0.37	0.62	0.63	0.97
LLaMA3	8B	✓	40k	0.36	0.62	0.60	0.83
Smol	135M	✓	70k	0.38	0.63	0.64	1.04
Smol	1.7B	✓	30k	0.38	0.62	0.63	1.07

Table 5

Results of the conditioning experiments. *Cond. Gen.* reports the quality of difficulty-conditioned generation, measured by difficulty classification accuracy (Acc; higher is better), mean squared error (MSE; lower is better), and validation cross-entropy loss (Loss; lower is better) on generated excerpts. *Diff.* reports the accuracy of the auxiliary difficulty prediction head on validation data. *deg.* indicates degenerate outputs for which metrics could not be computed.

Setup	Cond. Gen.			Diff.
	Acc (%)	MSE	Loss	Acc (%)
<i>Smol (135M) baseline</i>				
diff	69.62	0.38	1.32	–
diff_cot	69.33	0.30	1.34	–
feats	deg.	deg.	1.28	–
feats_cot	64.34	0.60	1.37	–
<i>Smol (135M) Auxiliary loss</i>				
diff	83.30	0.22	1.30	97.24
diff_cot	92.88	0.09	1.47	96.98
feats	deg.	deg.	1.05	96.42
feats_cot	67.17	0.54	1.39	97.25
<i>LLaMA 3 (1B) Auxiliary loss</i>				
diff	87.21	0.24	1.01	91.22
diff_cot	92.01	0.16	1.10	92.14
feats	51.57	0.92	1.02	54.88
feats_cot	64.12	0.60	0.97	90.35

benefit of using pre-trained language models. Among smaller models, Smol 135M generalizes better than GPT-2 and converges faster (70k vs. 120k), while Smol 1.7B converges in just 30k steps but underperforms compared to the LLaMA variants. LLaMA 1B, trained for 50k steps, also exceeds all smaller models.

Interestingly, larger models show better generalization in the *internal J-Pop* dataset, which is more out of distribution compared to the classical training data, suggesting higher robustness to domain shift. Despite this, Smol 135M achieves competitive results on all benchmarks. Moreover, it is based on an efficient architecture, which enables efficient deployment on CPUs and consumer GPUs. In particular, inference on a laptop GPU (e.g., 3050) takes <30s per piece, which is suitable for sight-reading educational use case. Therefore, we opted to employ Smol 135M as our minimalistic benchmark model and LLaMA 1B as our superior performance reference model in our comparative study. We refrained from using a larger model in order to maintain efficiency and balance resource consumption with performance outcomes.

5.2. Conditioning

We evaluated the effectiveness of difficulty conditioning strategies and the auxiliary classification objective described in Sections 3.3 and 3.4. Table 5 presents a comprehensive comparison in multiple setups. These include a 135M parameter model (*Smol*) trained with and without the auxiliary loss, and a larger LLaMA 3 model of 1B parame-

ters, always using the four prompting strategies introduced earlier: *diff*, *diff_cot*, *feats*, and *feats_cot* (see Table 2).

Each setup is evaluated along two axes: the quality of difficulty-conditioned generation and the accuracy of the auxiliary difficulty classification task. For the former, we compute the classification accuracy of the difficulty level and the mean squared error (MSE) of 1000 generated conditioned sequences, and the standard autoregressive cross-entropy loss in PDMX Val. For the latter, we report the prediction accuracy of the auxiliary classification head in PDMX Val. In cases where the model degenerated or the metric was not applicable, we report this explicitly.

The results show that using only the autoregressive loss (top block, Table 5) leads to relatively poor conditioning. For example, when using the *diff_cot* strategy without auxiliary loss, it reaches only 69.33% precision, while the *feats* strategy even degenerates. This supports the hypothesis of partial conditioning collapse, indicating that cross-entropy loss alone is suboptimal for difficulty-based control. The signal is too weak to effectively help minimize the loss of the next token prediction.

In contrast, when the auxiliary difficulty classification loss is added (middle block, Table 5), we observe large improvements across most metrics. With the *diff* prompt, generation accuracy improves from 69.62% to 83.30%, and with the more descriptive *diff_cot* prompt, it reaches 92.88%—the highest among all configurations. This supports the idea that chain-of-thought prompting helps models interpret difficulty intent more effectively. It also shows that the auxiliary head helps, improving both generation accuracy and feature alignment (MSE: 0.09). Furthermore, the classification accuracy of the auxiliary head is consistently above 96%, indicating that the model recognizes the difficulty of the generated sequence.

We observe that feature-based prompts (*feats*, *feats_cot*) consistently underperform compared to explicit difficulty-label prompts. While the extracted features are correlated with difficulty, they form a multidimensional and indirect control signal that appears harder for the model to exploit reliably during generation. In contrast, difficulty labels provide a simpler and more stable conditioning signal. From a pedagogical perspective, difficulty level is also the quantity of interest for the end user, as students and teachers typically reason in terms of *easy*, *medium*, or *advanced*, rather than low-level descriptors such as pitch entropy. For this reason, we treat difficulty-level conditioning as the primary control mechanism and report feature-based prompting as an exploratory baseline.

In the LLaMA 3 model (bottom block, Table 5), we observe strong generation performance even when using feature-based prompts (*feats_cot*), showing that scaling up the model helps overcome the limitations of less explicit conditioning. The model reaches a generation accuracy of 64% and a difficulty classification accuracy of 90.35%, confirming that larger models are more robust to weaker signals. The *diff_cot* experiment confirms our hypothesis that the chain-of-thought strategy improves performance. However, the results are similar to those of the Smol version, although with a better validation loss. The *feats* experiment did not degenerate, but shows very unstable behavior and poor results, highlighting the importance and effectiveness of the chain-of-thought strategy.



Fig. 5. Three 16-bar exercises generated by the LLaMA 3 Auxiliary Loss Model Diff (1B) at different target difficulty levels. The two excerpts on the left correspond to easy and medium levels, while the excerpt on the right illustrates an advanced example, showing increased rhythmic density and technical demands.

We observe that larger models achieve consistently lower generation loss but perform worse in the auxiliary difficulty prediction task. This pattern appears in several setups, including *diff*, *diff_cot*, and *feats_cot*. For example, in *diff*, the LLaMA 3 model reaches a generation loss of 1.01 but only 91.22% accuracy in difficulty prediction, while the smaller model achieves higher accuracy (97.24%) despite a higher loss (1.30). A similar trend is observed in *diff_cot* and *feats_cot*. Including the auxiliary head improves controllability and reduces conditioning collapse, even though its gradients are detached from the language model parameters. Previous studies in conditional text generation (Lapacz et al., 2025; Li et al., 2020; Yu et al., 2024) have shown that auxiliary tasks can indirectly shape internal representations, since the model must maintain hidden states that are simultaneously predictive of the next token and separable across difficulty classes. We hypothesize that, in our case, this auxiliary head acts as a regularizer that biases shared representations toward encoding difficulty-relevant features, thereby reinforcing the effect of conditioning. Future work will investigate this alignment more directly through representation analyses.

We also conducted experiments replacing the natural-language descriptions in *diff_cot* prompts with randomized text of the same length. Using the smaller *Smol-v2* model, this caused the task to degenerate, showing that semantic content is crucial. Moreover, although *feats_cot* prompts are longer than *diff_cot*, the latter performs better, suggesting that the gains come from linguistic priors and semantic alignment rather than prompt length. Finally, we tested the auxiliary objective without gradient detachment. In this case, the classification head quickly overfit by exploiting shortcuts from the prompt tokens, and controllability did not improve.

In general, these results confirm the benefits of combining explicit prompting strategies with a weak supervision signal from an auxiliary task. In particular, chain-of-thought formulations and the auxiliary classification head reinforce each other, improving the alignment between generated music and the intended difficulty level. In particular, larger models such as LLaMA 3 show strong performance even with the same scarce training data, suggesting that scaling up the model can compensate for limited supervision and weaker prompt structures. However, more research is needed to understand how to incorporate the composite signals in *feats* and *feats_cot* more effectively into the conditioning process.

5.3. Qualitative analysis of difficulty control

We analyze three complete exercises that were randomly generated by the LLaMA 3 Auxiliary Loss Model Diff (1B). Fig. 5 shows these ex-

amples. The discussion in this section focuses on their musical aspects. The next section, Section 6, reports a larger and independent evaluation conducted with independent music experts.

The first “easy” example has nine bars in E minor and 4/4 meter. The left-hand side uses only three pitch classes: E (I/tonic), B (V/dominant), and C (VI/submediant). A simple fingering works without any change of hand position: B with finger 4, E with finger 1, and C with finger 3. No cross-finger movement or lateral movement is required. The right hand is slightly more demanding. The rhythm includes anacrusis and some tied notes, and the line needs a few small hand shifts. This makes a good starting exercise for sight reading practice.

In the next “medium” example, the level increases. The main key is F minor, and there are many flattened chords that alternate between the two hands. Although the rhythmic figures are half- and quarter-notes, the hand positions are less comfortable than in the previous piece, and harmonies outside the key appear at several points. These characteristics make the exercise more challenging for a student than the previous exercise.

The last “advanced” example is clearly more difficult. The rhythm is denser, and there are many ascending and descending arpeggios, in the right hand. The main melody is played by the left hand, while the right hand performs a more virtuosic texture, in G minor. Accidentals and short modulations appear after bar 6. This exercise is more demanding than the previous ones; in particular, it has greater technical and musical complexity.

In general, the three examples show a gradual and coherent increase in difficulty. The material demonstrates that the system can generate musically consistent exercises suitable for different skill levels. Full scores and additional examples are available on the companion page.

6. User study

We conducted a user study with five pianists. All participants had formal piano training and between 25 and 40 years of experience (average: 35 years). Each pianist self-reported strong sight-reading skills, indicating that they could learn and perform a simple one-page piece (i.e., with difficulty comparable to Burgmüller’s *25 Etudes*) with less than one hour of practice. They also expressed confidence in their ability to estimate the performance difficulty of a piece by examining the score alone. Expertise was prioritized over participant quantity, as difficulty assessment requires deep pedagogical and performance knowledge. Recruiting such profiles is challenging, and involving five expert pianists already provides substantial value.

Table 6

Mean expert ratings for human-written excerpts (*Oracle*) and generated excerpts (*Proposed*) across three target difficulty levels. Ratings are reported for readability, naturalness, and playability on a five-point Likert scale, averaged across participants.

	Readability	Naturalness	Playability
<i>Oracle</i> /Easy	4.60	3.90	4.20
<i>Oracle</i> /Medium	2.80	3.80	3.10
<i>Oracle</i> /Advanced	2.10	3.60	2.40
<i>Proposed</i> /Easy	4.40	4.30	4.20
<i>Proposed</i> /Medium	3.20	3.90	3.80
<i>Proposed</i> /Advanced	3.60	3.90	3.50

6.1. Experimental conditions

We prepared twelve musical excerpts, each consisting of 16 measures. Six excerpts were sampled from the ground truth dataset, with two pieces for each of the three target difficulty levels (denoted as *Oracle*). Likewise, six excerpts were generated using the LLaMA 3 Auxiliary Loss Model Diff (1B), again with two pieces for each target difficulty level (denoted as *Proposed*).

Each participant was instructed to sight-read all excerpts. Participants were informed of the general purpose of the study but were not told how many generation methods were compared, nor whether a given score was human-written or machine-generated. The excerpts were presented in random order, and neither the generation method nor the target difficulty level was disclosed.

After each performance, participants were asked to rate three aspects of the score using a five-point Likert scale.

The evaluation criteria were designed to capture complementary aspects of sight-reading performance: readability reflects the visual and cognitive ease of parsing the notation at first sight, naturalness captures musical coherence and idiomatic writing independently of layout, and playability reflects the technical effort required to execute the score after learning.

- 1. Readability:** The cognitive ease of sight-reading the score. Ratings range from 1 (requires substantial time to study the score) to 5 (very easy to read). Participants were instructed to consider notation quality as part of this assessment.
- 2. Naturalness:** The degree to which the music feels musically coherent and idiomatic. Ratings range from 1 (musically incoherent) to 5 (entirely natural). As excerpts were intentionally limited to sixteen measures, participants were asked to assess naturalness with this design choice in mind.
- 3. Playability:** The technical ease of performing the piece after learning it, ranging from 1 (difficult) to 5 (easy).

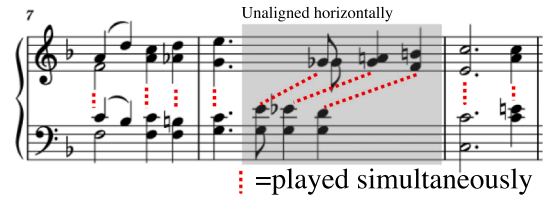
For each criterion, ratings were collected independently for every excerpt and participant. Reported results were obtained by averaging scores across participants for each condition and difficulty level. These ratings are intended as comparative expert judgments rather than as a psychometric evaluation of perceptual constructs.

Participants were also encouraged to provide additional free-form comments on each excerpt.

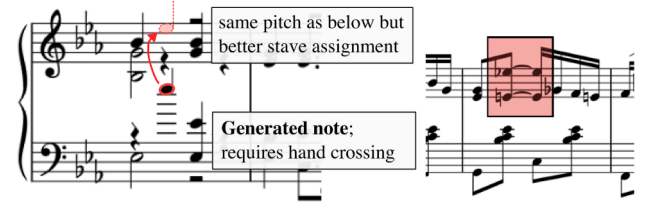
6.2. Results and discussions

6.2.1. Comparison of the questionnaires

Table 6 presents the ratings for the oracle and the proposed methods across the three difficulty levels. The playability ratings correspond inversely to the difficulty levels of the score. This indicates that scores conditioned to be more difficult are indeed perceived as more difficult to play – even when the performer has learned the piece. Since the playability tends to be lower for the oracle as the difficulty increases, the proposed method creates easy scores for a given difficulty level.



(a) Improper alignment.



(b) Unnatural voice crossing.

(c) Unnatural harmony

Fig. 6. Examples of notation and musical issues identified by the expert pianists during the user study, including alignment problems, unnatural voice assignment, and harmonically awkward passages.

Looking at the readability ratings, the proposed method seems to create pieces that are easier to sight-read than the oracle. The readability can be higher if the score is musically natural or the notation can be understood with little cognitive burden, but the consistently higher naturalness ratings suggest the former. The proposed method thus produces music that adheres more closely to familiar musical idioms than the oracle, making it easier to interpret and perform. Additionally, the higher playability ratings of the proposed method compared to the oracle suggest that the proposed method often generates simpler compositions.

These results show that our method can generate pieces (1) whose difficulty controls are qualitatively noticeable, (2) make musical sense, and (3) are playable. Surprisingly, the proposed method generated scores that rated better than the oracle. Although the underlying cause is unclear and requires further research, we observed that the oracle pieces tended to be more complex, as supported by the lower playability ratings, making proper rendering of MusicXML more difficult in music notation software. This might contribute to the oracle scores being more technically demanding, while having notation quirks.

6.2.2. Analysis of the comments

The participants left comments about parts that were difficult to read or play. The comments were written in a non-English language and translated into English in the subsequent discussions. In general, for each piece, the pianists tended to make similar remarks about specific measures rather than expressing concerns about the overall structure or musicality of the entire piece. This suggests that each composition contained a few markedly problematic points, rather than being uniformly unnatural or unplayable throughout. We show some of the commonly mentioned spots in Fig. 6.

The comments showed that the notation quality is a significant obstacle to sight-readability for both the oracle and the proposed method. A common and significant reported problem was the improper horizontal alignment, which increases cognitive load during sight-reading. Pianists commented that "overall, the bars are narrow and somewhat hard to read (proposed)," "the vertical are not aligned, making it hard to read (oracle, Fig. 6a)," and "the beats of right and left hands are not aligned (oracle)." Furthermore, the voicing, the assignment of a note to one of the two staves of the piano score, and its beam direction were another concern across both methods. Pianists noted that "left hand overlaps unnaturally with the right hand's registers (oracle)," "I get the impression that voice crossing was not intentionally considered (oracle)," and "having to play this note on left hand is strange (proposed, see Fig. 6b)." These observations and

the ratings suggest that the generated scores had some parts with confusing notation.

In addition to readability, the participants reported some musical and playability challenges. For example, the oracle included passages that *"have large leaps in a short interval,"* making them difficult to execute. The proposed method also contained passages that require the participant *"to look at [the] hand because the next note jumps too far"* or require *"unnatural hand shape [that] makes mistakes likely."* In addition, the proposed method occasionally contained musically unnatural elements, such as parts whose *"chord progression didn't feel right at first,"* or contained note combinations that *"seem too harsh for this kind of piece."* (Fig. 6c)

These suggest that the proposed method generated natural and playable music except for some unnatural leaps or pitch intervals. More importantly, confusing notation confused the participants regardless of whether the data was generated or written by a human.

7. Pedagogical implications

The proposed approach supports common practices in sight-reading pedagogy, where learners are regularly exposed to short, unseen excerpts at an appropriate difficulty level. Improved difficulty control enables teachers to generate exercises that match a student current ability and to increase complexity gradually, which is central to effective progression in reading fluency.

The expert evaluation suggests that the generated scores can serve as supplementary practice material in real teaching settings. Comparable ratings to human-written exercises in readability and naturalness indicate that the material is interpretable at first sight and musically plausible, which are essential properties for sight-reading practice in lessons, examinations, and self-study routines.

Qualitative feedback also highlights limitations that are pedagogically relevant. Issues such as horizontal spacing, staff assignment, or occasional awkward gestures can increase cognitive load during reading. This suggests that automatic post-processing in notation software, or additional generation constraints, may be necessary to ensure consistent classroom usability.

Overall, the system is best viewed as a tool to support music educators rather than replace them. By enabling interactive control and rapid generation of level-appropriate material, it can reduce preparation time and expand available practice resources, while teachers remain responsible for selecting, adjusting, and sequencing exercises according to individual learner needs.

8. Ethical statement

This work proposes a system to support piano teachers in creating personalized sight-reading exercises with controllable difficulty. The goal is twofold: to assist educators by automating repetitive tasks, such as generating varied, level-appropriate material, and to enable students to practice independently with exercises tailored to their skill level.

Although one might be concerned that such a system could negatively affect music teaching jobs, we believe that addressing the proposed task does not threaten the role of educators whose responsibilities extend well beyond the compilation of sight-reading exercises. Rather, this technology should be viewed as a supportive tool that helps teachers in their practice.

All training data used in this work come from open and copyright-free sources to ensure compliance with intellectual property rights. We also release the code, trained models, and evaluation resources to promote transparency and reproducibility, contributing to open science and enabling further research on personalized music education and difficulty-conditioned score generation.

9. Conclusion

This paper presents a method for generating piano exercises with controllable difficulty levels, targeting sight-reading practice. Our contributions are: first, we generate exercises directly in MusicXML format, ensuring scores are immediately usable in educational contexts; second, we introduce an auxiliary objective for difficulty prediction that improves adherence to difficulty conditioning and enables more reliable difficulty-aware generation. This work addresses a practical and previously underexplored problem in music education, where realistic piano scores for human reading are required rather than audio-oriented symbolic formats.

Experiments show that the proposed approach substantially improves difficulty control, and expert pianists rated the generated scores as comparable to human-written material in readability and naturalness. Although playability and readability were not explicitly supervised, these properties emerge from the data and the learning objective.

To clarify the scope of our evaluation, we note that direct comparison with existing symbolic music generation systems is not appropriate because most prior work targets MIDI, ABC, or audio-oriented representations, whereas our approach generates fully notated scores in MusicXML needed for sight-reading pedagogy. Conversion between symbolic formats is not lossless and may alter notation structure relevant to our evaluation. Furthermore, we are not aware of an openly available next-token generative model trained on large-scale MusicXML corpora under comparable conditions. Accordingly, our self-supervised initial MusicXML generative model is positioned as a baseline within this representation setting, supporting future research on MusicXML-based and education-oriented generative systems.

Although synthetic difficulty labels may introduce some noise, collecting large-scale expert annotations for generative training is logistically and financially prohibitive. Nevertheless, the results demonstrate that scalable and reliable control of global properties can be achieved by combining interpretable pedagogical descriptors with lightweight auxiliary objectives. This finding is relevant beyond the specific application, as it addresses conditioning collapse in generative models.

Generative approaches are particularly well suited to sight-reading practice, where real examinations rely on unseen material. The system is designed to support music educators by enabling interactive control at generation time without retraining. Overall, this work contributes an interdisciplinary approach at the intersection of education, art, and artificial intelligence, and provides a general strategy for controllable symbolic music generation.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT (OpenAI) to improve the readability and clarity of the manuscript. After using this tool, the authors carefully reviewed and edited the content to ensure accuracy and appropriateness, and they take full responsibility for the content of the published article.

CRediT authorship contribution statement

Pedro Ramoneda: Writing - original draft, Methodology, Investigation, Data curation; **Masahiro Suzuki:** Writing - review & editing, Methodology, Investigation, Data curation; **Akira Maezawa:** Writing - review & editing, Validation, Methodology; **Xavier Serra:** Writing - review & editing, Supervision.

Data availability

No data was used for the research described in the article.

Declaration of competing interest

Part of this research was conducted during a Yamaha Global Research Internship. However, the study is not related to any Yamaha product, and the authors declare no competing financial or personal interests.

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References

- Abed, S. A., & Khalil, Z. H. (2023). Reliability Allocation in Complex Systems Using Fuzzy Logic Modules. *Babylonian Journal of Mathematics*, 2023, (pp. 78–83). <https://doi.org/10.58496/BJM/2023/015>.
- Abed, S. A., Ghassan, M., Latef, S. Q., & Hassan, H. S. (2025). Reliability-based design optimization using differential-algebraic equations. *Iraqi Journal for Computer Science and Mathematics*.
- Allal, L. B., Lozhkov, A., Bakouch, E., Blázquez, G. M., Penedo, G., Tunstall, L., Marafioti, A., Kydlíček, H., Lajarin, A. P. et al. (2024). SmoLM2: When smol goes big – data-centric training of a small language model. [arXiv:2502.02737](https://arxiv.org/abs/2502.02737).
- Associated Board of the Royal Schools of Music (2024). Piano syllabus 2023 & 2024: Sight-reading requirements. Accessed: 2025-04-05 <https://www.abrsm.org>.
- Aydam, Z. M., & El Abbadi, N. K. (2025). Automated video colorization techniques for enhanced visual realism and computational efficiency. *Mesopotamian Journal of Computer Science*, 2025, (pp. Article 903).
- Bovin, A. (2021). Research shows software improves students' sight-reading skills. NAfME (National Association for Music Education) Blog. Accessed: 2026-01-10. URL: <https://nafme.org/research-shows-software-improves-students-sight-reading-skills>.
- C, E., Imam, A., Balarabe, M., & O, O. O. (2025). Graph-theoretic characterizations of quasi-idempotents in full order-preserving transformation semigroup. *Babylonian Journal of Mathematics*, 2025, (pp. 88–91).
- Cheng, C. C., Hu, D. J., & Saul, L. K. (2008). Nonnegative matrix factorization for real time musical analysis and sight-reading evaluation. In *Proceedings of the IEEE international conference on acoustics, speech and signal processing (ICASSP)*. Las Vegas, NV: IEEE.
- Chiu, S.-C., & Chen, M.-S. (2012). A study on difficulty level recognition of piano sheet music. In *Proc. of the IEEE int. symposium on multimedia (ism)*, Irvin, USA.
- Cook, N. (1999). Analysing performance and performing analysis. In N. Cook, & M. Everist (Eds.), *Rethinking music* (pp. 239–261). Oxford University Press.
- Dathathri, S., Madotto, A., Lan, J., Hung, J., Frank, E., Molino, P., Yosinski, J., & Liu, R. (2019). Plug and play language models: A simple approach to controlled text generation. [arXiv:1912.02164](https://arxiv.org/abs/1912.02164).
- Elliott, C. A. (1982). The relationships among instrumental sight-reading ability and seven selected predictor variables. *Journal of Research in Music Education*, 30(1), 5–14.
- Farley, A. L. P. (2014). The relationship between musicians' internal pulse and rhythmic sight-reading. University of Washington.
- Good, M. (2021). MusicXML version 4.0. <https://www.w3.org/2021/06/musicxml40/>. Accessed: 2025-07-10.
- Gover, M., & Zewi, O. (2022). amusic translation: Generating piano arrangements in different playing levels. In *Proceedings of the 23rd international society for music information retrieval conference (ISMIR)*, Bengaluru, India (pp. 1–9).
- Grattafiori, A., Dubey, A., Jauhri, A., Pandey, A., Kadian, A., Al-Dahle, A., Letman, A., Mathur, A., Schelten, A., Vaughan, A. et al. (2024). The llama 3 herd of models. [arXiv:2407.21783](https://arxiv.org/abs/2407.21783).
- Hayward, M., & Eastlund Gromko, J. (2009). Relationships among music sight-reading and technical proficiency, spatial visualization, and aural discrimination. *Journal of Research in Music Education*, 57(1), 26–36.
- Hsiao, W.-Y., Liu, J.-Y., Yeh, Y.-C., & Yang, Y.-H. (2021). Compound word transformer: Learning to compose full-song music over dynamic directed hypergraphs. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 178–186). AAAI Press (vol. 35).
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2022a). LoRA: Low-rank adaptation of large language models. In *Proceedings of the international conference on learning representations (ICLR)*.
- Hu, Z., Ma, X., Liu, Y., Chen, G., & Liu, Y. (2022b). The beauty of repetition in machine composition scenarios. In *Proceedings of the 30th ACM international conference on multimedia (MM '22)* (pp. 1223–1231). ACM.
- Huang, C.-H., & Lerch, A. (2019). On the automatic assessment of piano sight-reading skills. In *Proceedings of the international society for music information retrieval conference (ISMIR)*, Delft, Netherlands.
- Huang, C.-Z. A., Vaswani, A., Uszkoreit, J., Simon, I., Hawthorne, C., Shazeer, N., Dai, A. M., Hoffman, M. D., Dinculescu, M., & Eck, D. (2019). Music transformer: Generating music with long-term structure. In *Proceedings of the international conference on learning representations (ICLR)*.
- Huang, Y.-S., & Yang, Y.-H. (2020). Pop music transformer: Beat-based modeling and generation of expressive pop piano compositions. In *Proceedings of the 28th ACM international conference on multimedia (MM '20)* (pp. 1180–1188). ACM.
- Huron, D. (1995). The humdrum toolkit: Software for music research. <http://www.humdrum.org/>. Accessed: 2025-04-05.
- John, G. H., & Langley, P. (1995). Estimating continuous distributions in Bayesian classifiers. In *Proceedings of the 11th conference on uncertainty in artificial intelligence* (pp. 338–345). Morgan Kaufmann Publishers Inc.
- Lapacz, W., Marczak, D., Szatkowski, F., & Trzcinski, T. (2025). Exploring the stability gap in continual learning: The role of the classification head. In *IEEE/CVF winter conference on applications of computer vision, WACV*.
- Lee, J., Vafa, Z., Carlini, N., Raffel, C., & Taori, R. (2024). Few-shot style-conditioned LLM text generation via latent space interpolation. In *Submitted to international conference on learning representations (ICLR)*.
- Li, X. L., & Liang, P. (2021). Prefix-tuning: Optimizing continuous prompts for generation. In C. Zong, F. Xia, W. Li, & R. Navigli (Eds.), *Proceedings of the 59th annual meeting of the association for computational linguistics and the 11th international joint conference on natural language processing (volume 1: Long papers)* (pp. 4582–4597). Online: Association for Computational Linguistics.
- Li, Y., Li, C., Zhang, Y., Li, X., Zheng, G., Carin, L., & Gao, J. (2020). Complementary auxiliary classifiers for label-conditional text generation. In *The thirty-fourth AAAI conference on artificial intelligence*.
- Liu, Y., Zhang, Y., Liu, L., Wang, Q., Zhao, W. X., & Wen, J.-R. (2024). Controllable text generation for large language models: A survey. [arXiv:2408.12599](https://arxiv.org/abs/2408.12599).
- Long, P., Novack, Z., Berg-Kirkpatrick, T., & McAuley, J. (2025). PDMX: A large-scale public domain musicxml dataset for symbolic music processing. In *ICASSP 2025 - 2025 IEEE international conference on acoustics, speech and signal processing (icassp)* (pp. 1–5).
- Ma, X., Wang, Y., Kan, M.-Y., & Lee, W. S. (2021). AI-lyricist: Generating music and vocabulary constrained lyrics. In *Proceedings of the 29th ACM international conference on multimedia (MM '21)* (pp. 1002–1011). ACM.
- Mijwil, M. M., Ali, G., & Sadıkoğlu, E. (2023). The evolving role of artificial intelligence in the future of distance learning: Exploring the next frontier. *Mesopotamian Journal of Computer Science*, 2023, (pp. Article 28).
- Mohammed, Z. A., Damak, M., Fadhel, F. S., & Althainah, H. S. (2025). Existence and uniqueness theorem of multi-dimensional integro-differential equations with fractional differintegrations. *Babylonian Journal of Mathematics*, 2025, (pp. 44–49).
- MuseScore Development Team (2024). MuseScore 4.5 nightly build. <https://musescore.org/en/nightly-builds>. Accessed: April 2025.
- Nakamura, E., Ono, N., & Sagayama, S. (2014). Merged-output HMM for piano fingering of both hands. In *Proc. of the 15th int. society for music information retrieval conf. (ISMIR)*, Taipei, Taiwan.
- Nakamura, E., & Sagayama, S. (2015). Automatic piano reduction from ensemble scores based on merged-output hidden markov model. In *Proc. of the 41st int. computer music conf. (ICMC)*, Denton, USA.
- Nakamura, E., & Yoshii, K. (2018). Statistical piano reduction controlling performance difficulty. *APSIPA Transactions on Signal and Information Processing*, 7, e13.
- Pasquier, P., Ens, J., Fradet, N., Triana, P., Rizzotti, D., Rolland, J.-B., & Safi, M. (2025). MIDI-GPT: A controllable generative model for computer-assisted multitrack music composition. In *Proceedings of the AAAI conference on artificial intelligence*.
- Pike, P. D., & Shoemaker, K. (2013). The effect of distance learning on acquisition of piano sight-reading skills. *Journal of Music, Technology & Education*, 6(2), 147–162.
- Ramoneda, P., Jeong, D., Eremenko, V., Tamer, N. C., Miron, M., & Serra, X. (2024a). Combining piano performance dimensions for score difficulty classification. *Expert Systems with Applications*, 238, 121776.
- Ramoneda, P., Lee, M., Jeong, D., Valero-Mas, J. J., & Serra, X. (2025b). Can audio reveal music performance difficulty? insights from the piano syllabus dataset. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 33, (pp. 1129–1141).
- Ramoneda, P., Valero-Mas, J. J., Jeong, D., & Serra, X. (2023). Predicting performance difficulty from piano sheet music images. In *Proceedings of the 24th international society for music information retrieval conference (ISMIR)*, Milan, Italy.
- Ramoneda, P. et al. (2024c). Towards explainable and interpretable musical difficulty estimation: A parameter-efficient approach. In *Proceedings of the 25th international society for music information retrieval conference (ISMIR)*, San Francisco, USA.
- Sébastien, V., Ralambondrainy, H., Sébastien, O., & Conrui, N. (2012). Score analyzer: Automatically determining scores difficulty level for instrumental e-learning. In *Proc. of the 13th int. society for music information retrieval conf. (ISMIR)*, Porto, Portugal.
- Skopek, O., Aralikkat, R., Gooding, S., & Carbune, V. (2023). Towards better evaluation of instruction-following: A case-study in summarization. [arXiv:2310.08394](https://arxiv.org/abs/2310.08394).
- Spencer, P. (2011). sight-reading. Entry in *Grove Music Online*, Oxford University Press. <https://doi.org/10.1093/acref/978019579037.013.6180>
- Sturm, B. L., Santos, J. F., Ben-Tal, O., & Korshunova, I. (2016). Music transcription modelling and composition using deep learning. [arXiv:1604.08723](https://arxiv.org/abs/1604.08723).
- Suzuki, M. (2021). Score transformer: Generating musical score from note-level representation. In *Proceedings of the 3rd ACM international conference on multimedia in asia (MMAAsia '21)* (pp. 1–7). ACM.
- Suzuki, M. (2023). Piano score rearrangement into multiple difficulty levels via notation-to-notation approach. *J. Audio Speech Music Proc.*, 2023(52), (pp. Article 52).
- Thickstun, J., Hall, D., Donahue, C., & Liang, P. (2024). Anticipatory music transformer. *Transactions on Machine Learning Research*, (pp. EBNJ33Fcl).

- Trinity College London (2021). Piano 2021-2023 syllabus: Sight-reading criteria. <https://www.trinitycollege.com>. Accessed: 2025-04-05.
- Wareham, D. E. (1967). The development and evaluation of objective criteria for grading band music into six levels of difficulty. Ph.D. thesis. The Pennsylvania State University.
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E. H., Le, Q., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *arXiv:2201.11903*.
- Wristen, B. (2005). Cognition and motor execution in piano sight-reading: A review of literature. *Update: Applications of research in music education*, 24(1), 44–56.
- Wu, S., Li, X., & Sun, M. (2023a). Chord-conditioned melody harmonization with controllable harmonicity. In *ICASSP 2023 - 2023 IEEE international conference on acoustics, speech and signal processing (icassp)* (pp. 1–5). IEEE.
- Wu, S., Li, X., Yu, F., & Sun, M. (2023b). TunesFormer: Forming irish tunes with control codes by bar patching. *arXiv:2301.02884*.
- Yamaha Corporation (2025). Piano grade 10-6 examination procedure: Sight playing. Accessed: 2025-04-05 https://usa.yamaha.com/education/yms/grade_examination/student_grade/piano/10-6/index.html.
- Yan, Y., & Duan, Z. (2024). Measure by measure: Measure-based automatic music composition with modern staff notation. *Transactions of the International Society for Music Information Retrieval*, 7(1), 228–245.
- Yu, X., Bai, C., He, H., Wang, C., & Li, X. (2024). Regularized conditional diffusion model for multi-task preference alignment. In *Advances in neural information processing systems 38 (neurIPS 2024)*.
- Zhang, H., Karystinaios, E., Dixon, S., Widmer, G., & Cancino-Chacón, C. E. (2023). Symbolic music representations for classification tasks: A systematic evaluation. In *Proceedings of the 24th international society for music information retrieval conference (ISMIR), Milan, Italy*.
- Zhu, L., Xu, Y., Zhu, Z., Bao, Y., & Kong, X. (2023). Fine-grained sentiment-controlled text generation approach based on pre-trained language model. *Applied Sciences*, 13(1), 264.